Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/23/2006
PI Name:	Ronney, Paul D. Sc.D.		
Project Title:	Thermal Transportation Based Microscale Propulsion and Power Generation Devices		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Dunn, Bruce (University of California at Los Angeles)		
Grant/Contract No.:	NNC04GA31G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this work we introduce a self-sustaining, self-pressurizing catalytic combustion-driven thermal transpiration pump and reactor. Thermal transpiration occurs in capillary tubes or porous material when (1) a temperature gradient is maintained across the tube or pore and (2) the tube or pore diameter is comparable to the mean free path of gas molecules. Under such conditions a pressure gradient is produced in the same direction as the temperature gradient. Thermal transpiration (thus pressurization) is accomplished using membranes made of nanoporous materials (aerogels and glass-fiber filters) that have pore sizes comparable to the mean free path of gas molecules and have very low thermal conductivity, thereby enabling temperature gradients to be sustained with minimal thermal power. Catalytic combustion of hydrogen-air mixtures on the downstream side of the membrane provided the thermal power for self-sustaining pumping of the reactants. The objective of this work is to determine whether pumping and propulsion based on this catalytic combustion driven		

Task Description:	thermal transpiration concept is feasible and if so to determine the optimal design and operating conditions for devices exploiting this concept. Due to recent advances in MEMS technologies, there is considerable interest in miniaturizing thermochemical systems for electrical power generation, gas chromatography, gas separation and chemical reforming, and propulsion for micro air vehicles. These systems all require gas pressurization and/or vacuum pumping systems that in turn require devices with moving parts that are difficult to miniaturize due to problems with friction losses, manufacturing tolerances, sealing, etc. The proposed device accomplishes pumping with no moving parts and using thermal rather than electrical energy as its feedstock.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed devices are applicable to portable electronic devices, microcombustors, microreactors and micro propulsion devices on earth as well as in space.
Task Progress:	By use of a novel tetrahedral configuration that resulted in minimal heat losses, catalytic combustion of propane-air mixtures on the downstream side of the membrane was able to provide the thermal power for self-sustaining pumping of the reactants. This proof-of-concept apparatus demonstrates the first-ever example of a gas pump with no moving parts that derives its energy for pumping from the gas itself with no external energy source. Transpiration properties of glass fiber discs were enhanced by coating the disc with either a traditional silica aerogel or a macroporous silica aerogel. These results show the advantages of using materials with designed microstructures. In particular, we have emphasized the development of macroporous aerogels in which the large pore volume is determined by templating methods. The macroporous aerogel material was far more effective than the traditional aerogel as the maximum flow rate of the macroporous aerogel coated discs was 11 mL/min at a temperature difference of 280°C. Unlike the uncoated glass fiber disc, the coated disc did not exhibit a decrease in flow rate at elevated temperatures. In addition, platinum catalytic particles were fabricated by the reduction of hydrogen hexachloroplatinate in ethylene glycol.
Bibliography Type:	Description: (Last Updated: 01/29/2007)
Abstracts for Journals and Proceedings	Ochoa F, Ronney PD. "A thermal transpiration-based self-pressurizing mesoscale combustor." 4th Joint U.S. Sections Meeting, Combustion Institute, Philadelphia, PA, March 2005. Proceedings, 4th Joint U.S. Sections Meeting, Combustion Institute, Philadelphia, PA, March 2005. , Mar-2005
Abstracts for Journals and Proceedings	Ochoa F, Ronney PD. "A thermal transpiration based self-pressurizing mesoscale combustor for power MEMS devices." 6th Power MEMS Conference, Berkeley, CA, Nov. 29-31, 2006. Conference proceedings, 6th Power MEMS Conference, Berkeley, CA, November 2006. , Nov-2006